

CS6660: MATHEMATICAL FOUNDATIONS OF DATA SCIENCE
(PROBABILITY)
PRACTICE PROBLEMS 03



1. Let $X, Y \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. Deduce the distribution of $Z = X + Y$.
Repeat the same exercise when $X, Y \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$ for some fixed $\lambda > 0$.
2. Suppose that X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} cy, & -1 \leq x \leq 1, 0 \leq y \leq |x|, \\ 0, & \text{otherwise.} \end{cases}$$

Determine $\mathbb{P}(\{|X - Y| > 0.5\})$.

3. Two numbers are drawn independently and uniformly from the unit interval $[0, 1]$.
The smaller of the two the numbers is known to be less than $1/3$.
What is the probability that the larger one is greater than $3/4$?
4. Let $X_1, X_2, \dots$ be independent random variables distributed uniformly on $(0, 1)$. Let N be defined as

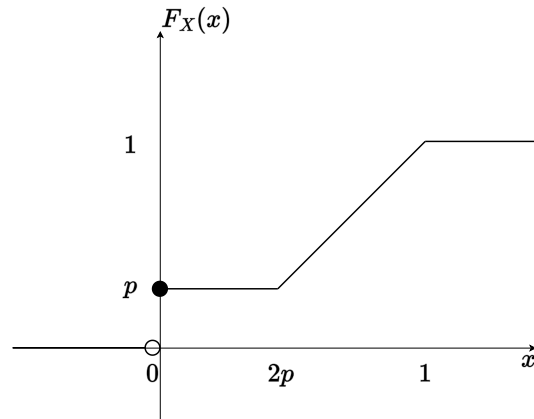
$$N = \min\{n \geq 1 : X_{n+1} > X_n\}.$$

For instance, if $X_1 = 0.5, X_2 = 0.45, X_3 = 0.425, X_4 = 0.95$, then $N = 4$.

Determine the PMF p_N of N .

Hint: Compute $\mathbb{P}(\{N > n\})$ for each $n \in \mathbb{N}$, and think how you can use this to compute $p_N(n) = \mathbb{P}(\{N = n\})$.

5. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$. Derive the marginal PDFs of $Y_n = \min\{X_1, \dots, X_n\}$ and $Z_n = \max\{X_1, \dots, X_n\}$.
6. Consider a random variable X with CDF F_X as depicted below for some fixed $0 < p < 1/2$.



Sketch the CDFs of $Y = \min\{X, p\}$ and $Z = \max\{X, p\}$, and compute $\mathbb{P}(\{Y \geq Z - 1\})$.